

~~Q1~~ There exists a rabbit who knows Calculus.

c) Every bird can fly.

Domain: The set of animals $B(x): x$ is a bird $F(x): x$ can Fly.

$\forall x (B(x) \rightarrow F(x))$ Negation: $\neg \forall x (B(x) \rightarrow F(x)) \equiv \exists x \neg (B(x) \rightarrow F(x))$
(By De Morgan's law of quantifiers) $\equiv \exists x \neg (\neg B(x) \vee F(x))$ (Conditional-disjunction equivalence) $\equiv \exists x \neg (\neg B(x) \wedge \neg F(x))$ (2nd De Morgan's law) $\equiv \exists x B(x) \wedge \neg F(x)$ (Double-negation law)

There exists a bird who cannot fly.

d) There is no dog that can talk.

Domain: The set of animals. $D(x): x$ is a Dog $T(x): x$ can Talk

$\forall x (D(x) \rightarrow \neg T(x))$ Negation: $\neg \forall x (D(x) \rightarrow \neg T(x)) \equiv \exists x \neg (D(x) \rightarrow \neg T(x))$
(De Morgan's law of quantifiers) $\equiv \exists x \neg (\neg D(x) \vee \neg T(x))$ (Conditional-disjunction equivalence) $\equiv \exists x \neg (\neg (D(x) \wedge T(x)))$ (1st De Morgan's law) $\equiv \exists x (D(x) \wedge T(x))$ (Double negation law)

There exists a dog that can talk.

e) There is no one in this class who knows French and Russian. ??

Domain: Students of the class $F(x): x$ knows French
 $R(x): x$ knows Russian

Negation: $\neg \forall x (\neg (F(x) \wedge R(x))) \equiv \exists x \neg (\neg (F(x) \wedge R(x)))$ (De Morgan's laws of quantifiers) $\equiv \exists x \neg (\neg F(x) \vee \neg R(x))$ (2nd De Morgan's law) $\equiv \exists x (F(x) \wedge R(x))$ (By Double-negation Law). There exists a student in the class who knows French or Russian.

35) Express the negation of each of these statements in terms of quantifiers without using the negation symbol.

a) $\forall x (x > 1)$ $\neg \forall x (x > 1) \equiv \exists x \neg (x > 1)$ (By De Morgan's laws of quantifiers) $\equiv \exists x (x \leq 1)$